Dr. D. Y. Patil Educational Federation's
Dr. D. Y. Patil College of Engineering & Innovation
Approved by AICTE, Recognized by Govt. of Maharashtra,
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Accredited by NAAC with "A" Grade

Department of Computer Engineering

Program name :- Artificial Intelligence and Data Science Class: S.Y. B.Tech Div:- B

Name of Student: Raj Nikam Batch:- B1 Course Name:- DF

PRN No :- 251111075__Roll No:- 22215 Sem :- 3rd Course :- BSFI (Online)

SMARTBRIDGE | SKILLWALLET

AI Loan Eligibility Checker: AI-Powered Loan Readiness & Financial Assistant

Project Description :-

AI Loan Eligibility Checker is an educational financial assistance web application designed to help users understand their loan readiness before considering a loan. The platform brings together loan eligibility analysis, credit score interpretation, EMI calculation and general financial guidance in a single user-friendly application.

The Loan Eligibility module allows users to enter important financial inputs such as age, monthly income, employment type, existing monthly EMI, credit score and requested loan amount. The system processes these inputs and provides an educational estimate of loan readiness along with relevant reasons and repayment information.

The application also provides a Credit Score Analyzer for understanding a manually entered score, an EMI Calculator for estimating monthly repayment, and an AI Financial Tips section for general educational guidance related to loans, debt, credit and saving.

The project is designed with a responsive and simple interface so that users can move between the four main tools easily. The current application is intended for educational demonstration and does not represent actual bank approval or connect to a credit bureau.

Technology & AI :-

Frontend: React.js / TypeScript / HTML / CSS

Build Tool: Vite

Financial Logic: Client-side loan eligibility and EMI calculation

AI / Guidance: AI Financial Tips module with safe educational fallback guidance

Technology & AI :-

Frontend: React.js / TypeScript / HTML / CSS

Build Tool: Vite

Financial Logic: Client-side loan eligibility and EMI calculation

AI / Guidance: AI Financial Tips module with safe educational fallback guidance

Version Control: Git / GitHub

Key Features :-

• Loan eligibility/readiness analysis based on user-provided financial inputs

• Age, monthly income, employment type and existing EMI input handling

• Credit score input and general credit-score range analysis

• Requested loan amount and repayment-related analysis

• Estimated EMI calculation

• Principal, total interest and total repayment display

• Interactive EMI visualization

• AI Financial Tips for general educational financial guidance

• Safe fallback guidance when live AI is not configured

• Responsive and user-friendly dashboard interface

• Separate pages/routes for Loan Eligibility, Credit Analyzer, EMI Calculator and AI Financial Tips

• Clear financial results and explanations for easier understanding

• No direct connection to or retrieval of data from any credit bureau

• Educational/demo-oriented financial analysis rather than actual bank approval

Scenarios :-

Scenario 1: A user opens the Loan Eligibility Checker and enters age, monthly income, employment type, existing EMI, credit score and requested loan amount. The system processes the provided information and displays an educational loan-readiness result.

Scenario 2: A user enters a credit score in the Credit Score Analyzer. The application interprets the score range and provides general educational information about what the score can mean.

Scenario 3: A user opens the EMI Calculator, enters a loan amount, annual interest rate and loan tenure. The system calculates the estimated monthly EMI and displays principal, total interest and total repayment information.

Scenario 4: A user visits AI Financial Tips and enters a general question about loans, debt, credit or saving. The application provides educational guidance and uses a safe fallback when a live AI service is not configured.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objectives of AI Loan Eligibility Checker, identify target users, finalize the four core financial tools and design the overall application workflow.

Milestone 2 – Environment Setup & Frontend Development

Set up the React.js and TypeScript project using Vite, organize routes and components, and create the responsive user interface.

Milestone 3 – Loan Eligibility & Financial Input Module

Develop the loan eligibility form, validate user inputs and implement the logic for analyzing income, existing EMI, credit score and requested loan amount.

Milestone 4 – Credit Analysis & EMI Calculation

Implement credit score analysis and EMI calculation, including estimated monthly EMI, principal, total interest and total repayment with visual presentation.

Milestone 5 – AI Financial Tips & User Experience

Develop the AI Financial Tips section, provide general educational guidance, implement safe fallback behavior and improve navigation and user experience across the application.

Milestone 6 – Testing, Optimization & Deployment

Test all financial calculations, forms, routes, result displays and responsive behavior. Fix errors, improve reliability and prepare the project for GitHub-based submission and demonstration.

Conclusion :-

AI Loan Eligibility Checker provides a simple and interactive way for users to understand important factors related to loan readiness. By combining loan eligibility analysis, credit score interpretation, EMI calculation and financial guidance, the application converts basic financial inputs into clear and easy-to-understand educational information.

The project demonstrates how modern frontend technologies can be used to build a practical financial assistance application with multiple tools in a single interface.

The project demonstrates how React.js, TypeScript, Vite, client-side financial logic and responsive UI design can be combined to develop an educational FinTech-oriented web application for loan readiness, credit awareness, EMI planning and financial guidance.

Expected Outcome :-

AI Loan Eligibility Checker aims to help users better understand their basic financial position before considering a loan. The platform provides an educational loan-readiness analysis, credit score interpretation, automated EMI calculations and general financial guidance in a single application.

The project will demonstrate practical expertise in React.js, TypeScript, Vite, frontend development, financial calculations, responsive UI design, client-side application logic, Git/GitHub version control and AI-oriented financial guidance, while providing a foundation for future integration with real-world financial services and secure backend systems.